Methods results and limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which evidence makes this a procedure, finding, limitation or proposed explanation.

Teach faithful interpretation without treating unstudied conditions as negative results.

Keep one habit: Model main roles → compute supplied denominators → inspect outdoor overclaim → assess reviewer overlap → preserve missing methods/evidence.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **method**s **result**s and **limitation**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A **result** depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of finding primary research

Reference example, not the only acceptable wording: Teacher explanation: Primary research presents a study's own **method**s and **result**s. A summary can help discovery but should not replace inspection of the original **evidence**.

Record title, authors, version, date, identifier and access location. Distinguish paper, code, data and commentary. Verify that a quoted claim belongs to the cited version.

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of a question that **evidence** can answer

Reference example, not the only acceptable wording: Teacher explanation: A research question identifies an outcome, comparison and conditions that **evidence** could answer. A broad aspiration needs narrowing before a study.

For fictional classroom retrieval, ask whether **method** B returns more relevant records than lexical baseline A on a reserved judged query set at cutoff3. Define eligible corpus and relevance rules before running.

Pictured end goal

Explain **method**s **result**s and **limitation**s using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Finding primary research

A question that evidence can answer

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. **Method**s state what was done; **result**s state observed outcomes; **limitation**s state restrictions on interpretation. Mixing them hides missing **evidence**.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and **limitation**s; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **method**s **result**s and **limitation**s

Apply the stated procedure to a new case with a traceable **result**

Identify an assumption or **limitation** and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach role distinctions with reasons and legitimate overlap

> A plausible explanation must remain separate from observed results.

Definitions and complete passage contract

Methods state what was done; **result**s state observed outcomes; **limitation**s state restrictions on interpretation. Mixing them hides missing **evidence**.

Classify each statement by role and link **result**s to the procedure and denominator. A proposed **mechanism** is an explanation to test, not an extra **observation**.

Statement role	Complete classroom distinction
Method	What was done: cases, versions, scoring, conditions and data collection
Result	What the supplied record reports, with its metric and denominator
Limitation	What that procedure/evidence does not establish or what uncertainty remains
Proposed explanation	A candidate mechanism to test; not an additional observed result
Main supplied passage	Used 20 fictional queries; 16 passed; real student use not studied
Main arithmetic	16/20 = 80% on those fictional queries
Missing details	Full pass rubric, case records and sampling protocol are not supplied by the short passage
Separate picture	Two frozen classifiers, same five labelled leaf photos, one camera, no outdoor photos
Picture result	A 3/5 = 60%; B 4/5 = 80%, difference 20 percentage points
Scope	Fictional reported evidence, not a new model experiment, classroom-learning study or outdoor deployment evaluation

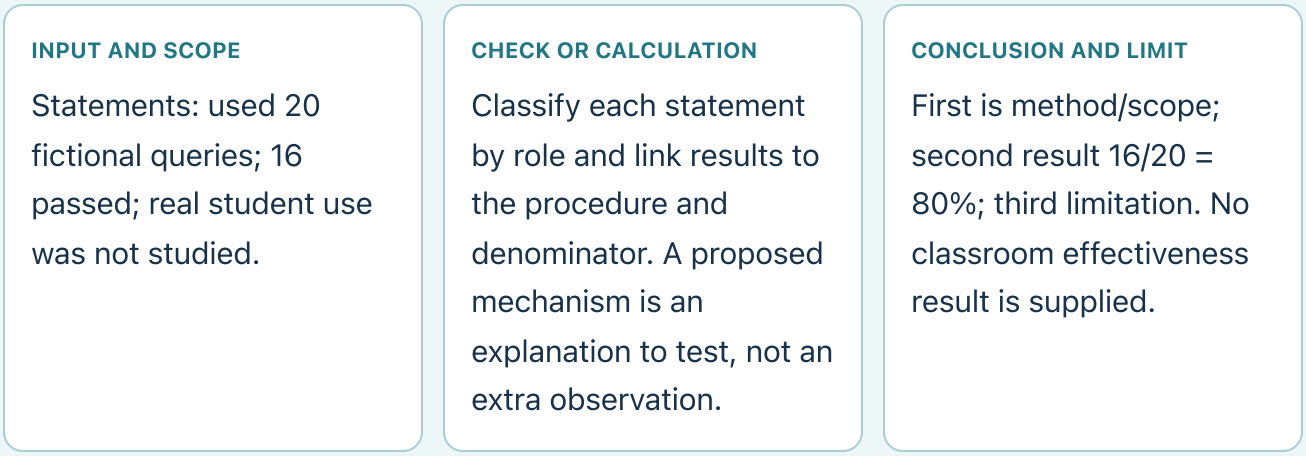
Main classification ledger and checked reasoning

Main supplied statement	Role and reason	Faithful use
Used twenty fictional queries	Method and scope: defines what was evaluated	Describe those query cases, not real learners
Sixteen passed	Reported result; needs total and pass rule	16/20 = 80% on the supplied cases
Real student use was not studied	Limitation and study-scope detail	No classroom effectiveness result supplied
“It worked because students learned faster”	Unsupported proposed explanation/outcome	Neither student use nor learning speed was measured
Full rubric and raw case identities absent	Evidence gap in the short summary	Ask for them; do not invent reproducibility details

Main twenty fictional queries is method/scope, sixteen passes yields 16/20 = 80% reported result, real student use absent is limitation; no learning result or mechanism supplied. Picture same five photos/labels and frozen classifiers is method; A 3/5=60%, B 4/5=80% is result; no outdoor cases limits generalization, not proof of outdoor failure.

Selected cutoff 5 is method; 12/15 relevant returned items=80% is reported result; one reviewer is a method detail with unassessed cross-reviewer agreement as a limitation. Additional independent review could assess agreement; do not claim labels are wrong or agreement measured. All passages are fictional and no model/learner experiment is run.

Methods results and limitations — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: First is **method**/scope; second **result** 16/20 = **80%**; third **limitation**. No classroom effectiveness **result** is supplied. Qualification: A **limitation** should constrain a claim rather than be hidden in unrelated prose. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Statements: used 20 fictional queries; 16 passed; real student use was not studied.

First is **method**/scope; second **result** 16/20 = **80%**; third **limitation**. No classroom effectiveness **result** is supplied.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Role overlap, reviewer and scope probes

Role-reading probe	Expected interpretation
Count with no denominator	Incomplete result statement; locate total and scoring
Same cases and labels	Method comparison condition, not the result itself
One reviewer assigned all labels	Method detail; agreement across reviewers not assessed
Single reviewer means labels must be wrong	Unsupported; limitation is not proof of error
No outdoor cases tested	Bounds interpretation; does not establish outdoor failure
A plausible mechanism explains a difference	Hypothesis, not an extra observation
One statement serves two roles	Name both with reasons rather than forcing a single label
Methods paragraph too brief to reproduce	Request missing details; do not invent settings
80% on fictional queries	No measured real-student learning effect follows

Separate five-photo classification and outdoor-evidence gap

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.

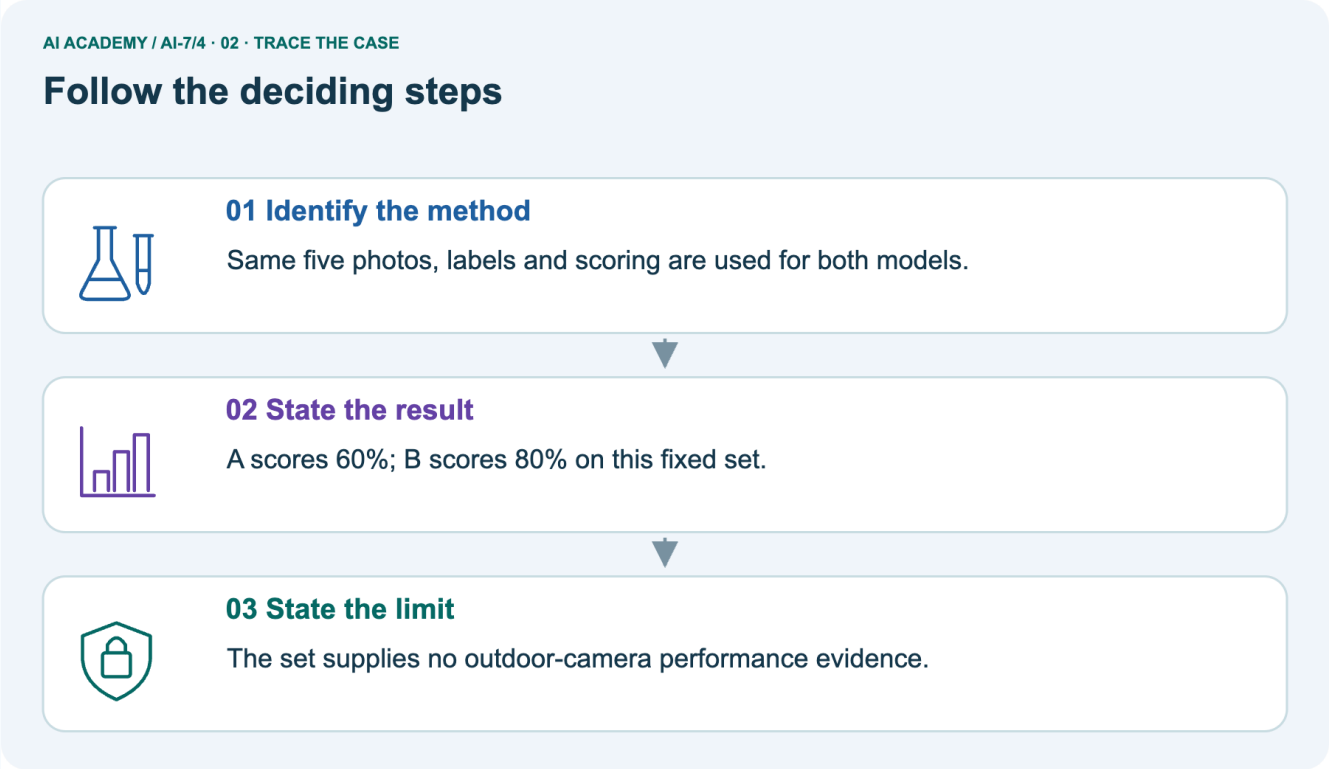
A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.

The picture is a separate five-photo evaluation. Method: two frozen classifiers use the same five labelled leaf photos and scoring, from one camera, with no outdoor photos. Result: A is correct on 3/5 = 60%, B on 4/5 = 80%, so B is higher by 20 percentage points on this fixed set. Limitation: outdoor conditions, other cameras and broader sampling are not evaluated.

“B is best for outdoor farming” goes beyond the available comparison. The five-photo result neither proves that B will generalize nor proves it will fail outdoors. It also does not identify an internal causal mechanism.

Separate picture statement	Role	Evidence-faithful wording
Two frozen classifiers; same five labelled photos, one camera	Method	State versions, same photos, labels and scoring conditions
A 3/5 and B 4/5	Reported result	A 60%, B 80%; B higher on these five cases
No outdoor photos	Method/scope detail and limitation	Outdoor performance unassessed
B best for all outdoor farming	Unsupported conclusion	Replace with the fixed-set comparison and explicit missing outdoor evidence

Picture trace and oversized-conclusion key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify the method	Same five photos, labels and scoring are used for both models.
2. State the result	A scores 60%; B scores 80% on this fixed set.
3. State the limit	The set supplies no outdoor-camera performance evidence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Methods tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

Five **observation**s cannot establish a stable ranking or causal **mechanism**; source and sampling assumptions remain important.

Change: The conclusion says B is best for outdoor farming.

Prediction: Reject that extension without outdoor **evidence**.

Reason: The reported conditions do not include the proposed use population.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Method	The procedure used to obtain observations
Result	A reported observation or calculated outcome
Limitation	A restriction on what results establish

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

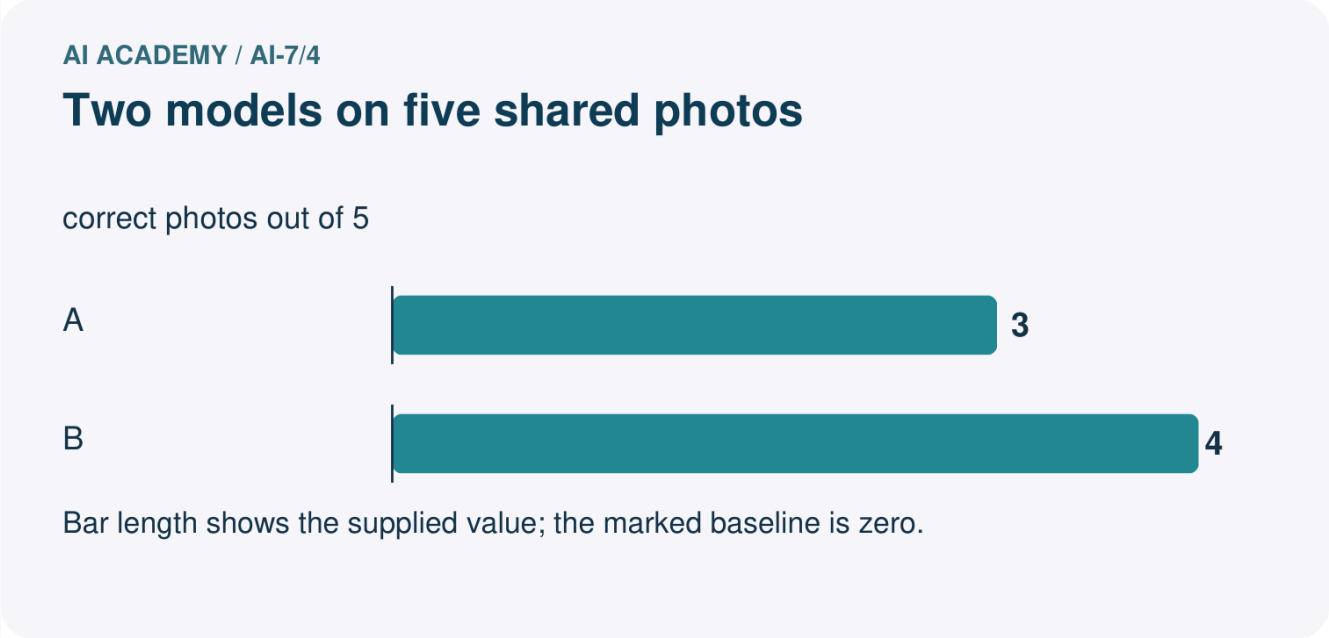
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Methods results and limitations

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/4

Methods results and limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the method

Same five photos, labels and scoring are used for both models.



2. State the result

A scores 60%; B scores 80% on this fixed set.



3. State the limit

The set supplies no outdoor-camera performance evidence.

CHANGE ONE THING

The conclusion says B is best for outdoor farming.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the **result**. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Statements: used 20 fictional queries; 16 passed; real student use was not studied. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: First is **method**/scope; second **result** 16/20 = **80%**; third **limitation**. No classroom effectiveness **result** is supplied.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Method**s **result**s and **limitation**s. Proposed explanation: “The worked **result** establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A **limitation** should constrain a claim rather than be hidden in unrelated prose.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the **result** establish that B is best for outdoor farming?

Same five photos, labels and scoring are used for both models. A scores **60%**; B scores **80%** on this fixed set. The set supplies no outdoor-camera performance **evidence**.

Changed case: Reject that extension without outdoor **evidence**. The reported conditions do not include the proposed use population.

Changed-case reasoning: Reject that extension without outdoor **evidence**. The reported conditions do not include the proposed use population.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and **result** to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its **result**; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/**result**/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main20-query procedure, 16/20 = 80% result and no-real-student limitation must stay connected. Picture fixed five-photo A 60/B 80 result does not establish outdoor best or a causal mechanism. Selected cutoff 5 is method; 12/15 = 80% is reported result; one reviewer is both a method detail and limitation because cross-reviewer agreement is unassessed.

Suggest an appropriate agreement check without inventing a measured value or declaring the labels wrong.

Diagnose the current misconceptions

The worked **result** establishes every future case: A **limitation** should constrain a claim rather than be hidden in unrelated prose.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Statements: fixed cutoff 5; 12/15 relevant returned items; judgments came from one reviewer. Classify.

Teacher explanation: Cutoff is **method**; 12/15=**80%** is a measured **result**; single-reviewer judgment is a **limitation** and **method** detail requiring agreement checks.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Method**s state what was done; **result**s state observed outcomes; **limitation**s state restrictions on interpretation. Mixing them hides missing **evidence**.

Classify each statement by role and link **result**s to the procedure and denominator. A proposed **mechanism** is an explanation to test, not an extra **observation**.

W2 Trace the worked case

Statements: used 20 fictional queries; 16 passed; real student use was not studied. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: First is **method**/scope; second **result** 16/20 = **80%**; third **limitation**. No classroom effectiveness **result** is supplied.

W3 Audit the limit

A report omits this qualification: A **limitation** should constrain a claim rather than be hidden in unrelated prose. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A **limitation** should constrain a claim rather than be hidden in unrelated prose.

State the observed **result** separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A **limitation** should constrain a claim rather than be hidden in unrelated prose.

The new case should test this particular **mechanism** rather than merely restate the original **result**.

▼ **Earlier wording, alternatives and picture transcripts**

Guided and exit questions

Guided: First is **method**/scope; second **result** 16/20 = **80%**; third **limitation**. No classroom effectiveness **result** is supplied. Qualification: A **limitation** should constrain a claim rather than be hidden in unrelated prose. Exit: accept a concrete variation with its predicted

outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Statements: fixed cutoff 5; 12/15 relevant returned items; judgments came from one reviewer. Classify.

Teacher explanation: Cutoff is **method**; 12/15=**80%** is a measured **result**; single-reviewer judgment is a **limitation** and **method** detail requiring agreement checks.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Cutoff is **method**; 12/15=**80%** is a measured **result**; single-reviewer judgment is a **limitation** and **method** detail requiring agreement checks. Require intermediate reasoning and the stated scope.

Fresh retry: Fixed seed; 7/10 success; no mobile evaluation. Expected reasoning: **Method**, **result70%**, **limitation** respectively. The seed does not remove the mobile **evidence** gap.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **method**s **result**s and **limitation**s **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **method**s **result**s and **limitation**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and **limitation**. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/**result**/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Replication and reproduction. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/5, ai-7/7, ai-7/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **method**s **result**s and **limitation**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Statements: used 20 fictional queries; 16 passed; real student use was not studied. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: First is **method**/scope; second **result** 16/20 = **80%**; third **limitation**. No classroom effectiveness **result** is supplied.

Check every learner before discussion: Use the worked case for **Method**s **result**s and **limitation**s. Proposed explanation: “The worked **result** establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A **limitation** should constrain a claim rather than be hidden in unrelated prose.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The conclusion says B is best for outdoor farming. Require a **prediction** and a reason before comparing. Expected **result**: Reject that extension without outdoor **evidence**.

Reason: The reported conditions do not include the proposed use population.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting **method**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and **limitation**s. Progress ticks and quiz selections alone do not establish mastery.

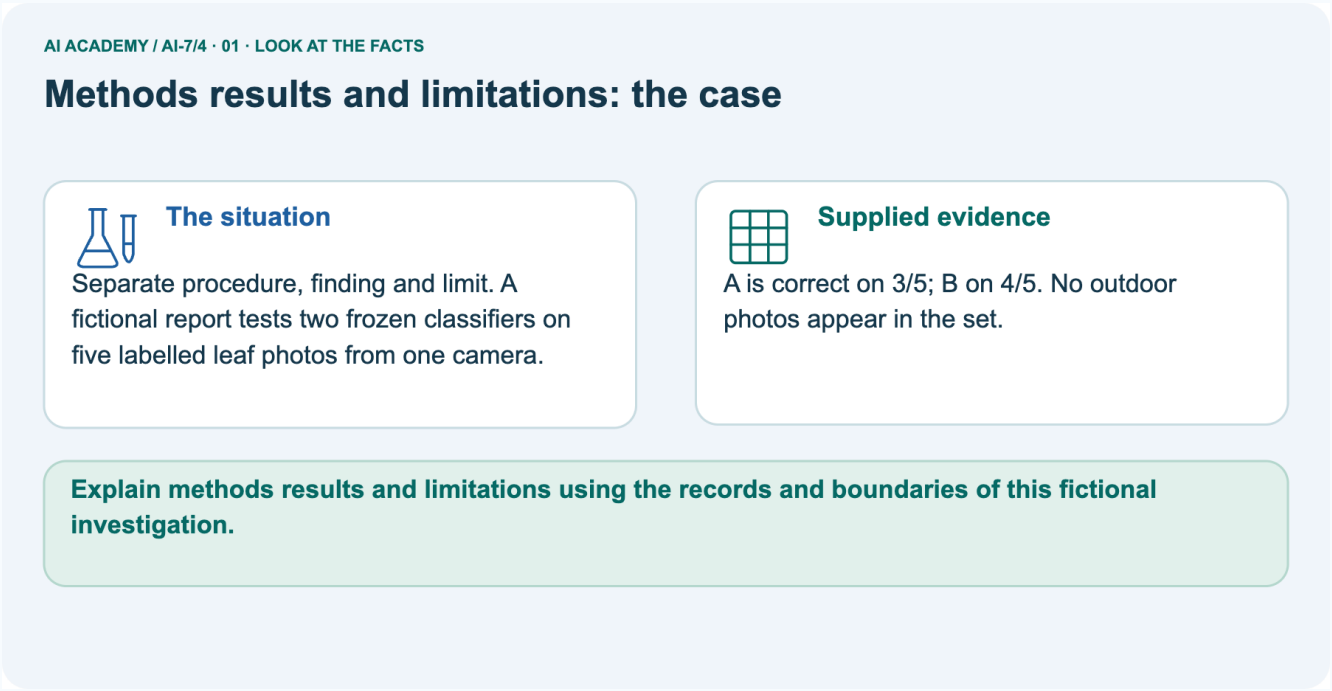
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Methods results and limitations: the case



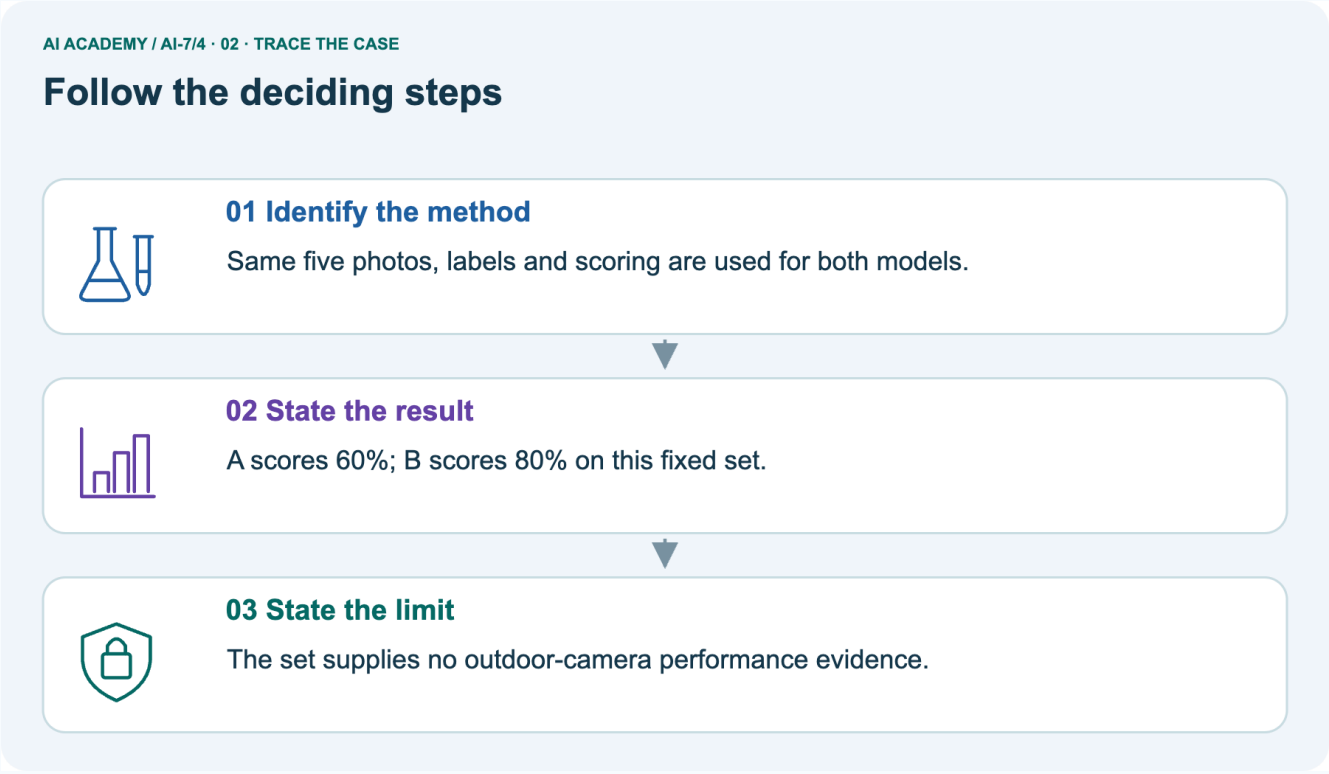
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera. Supplied facts: A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



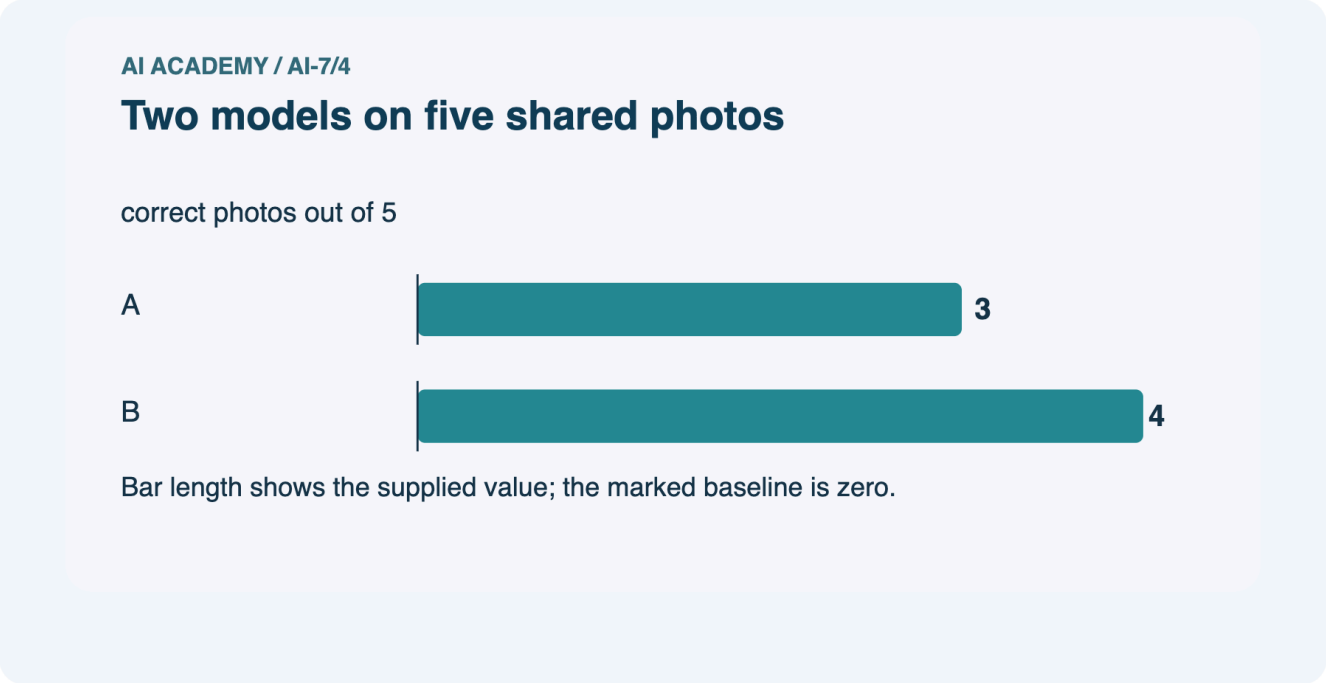
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the **method**: Same five photos, labels and scoring are used for both models. → State the **result**: A scores **60%**; B scores **80%** on this fixed set. → State the limit: The set supplies no outdoor-camera performance **evidence**.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two models on five shared photos



Follow the labelled picture with the matching explanation. Open the vector diagram

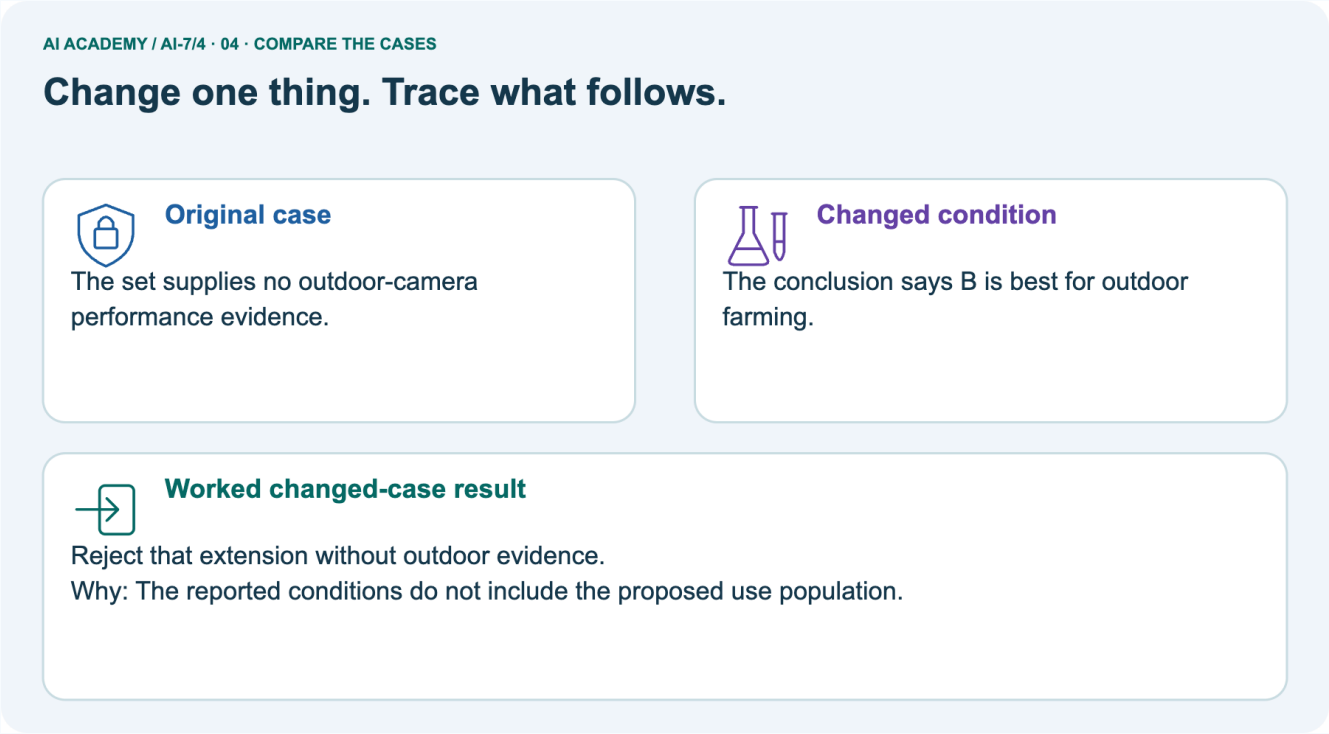
Two models on five shared photos. Units: correct photos out of 5. A: 3. B: 4.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model **result**s.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The set supplies no outdoor-camera performance **evidence**. Changed condition: The conclusion says B is best for outdoor farming. Expected **result**: Reject that extension without outdoor **evidence**.

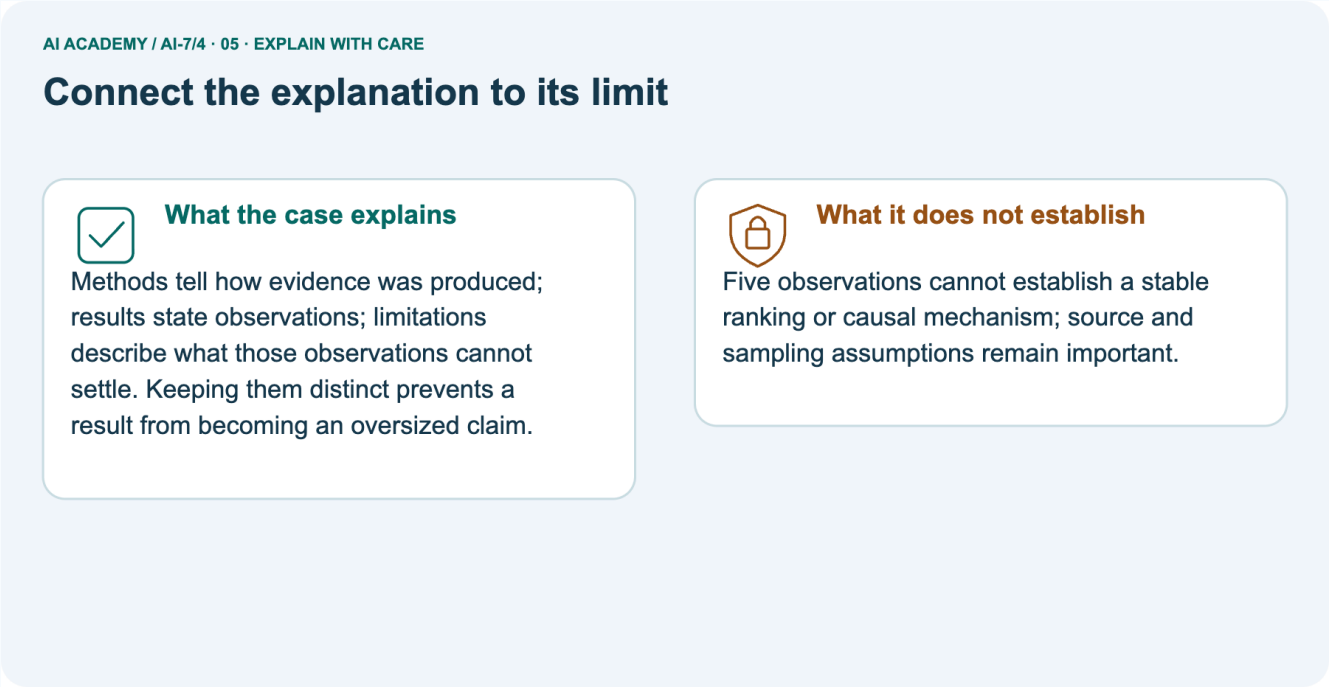
Reason: The reported conditions do not include the proposed use population.

Predict before checking the worked changed-case **result**. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case **result**. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The reported conditions do not include the proposed use population.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Methods tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

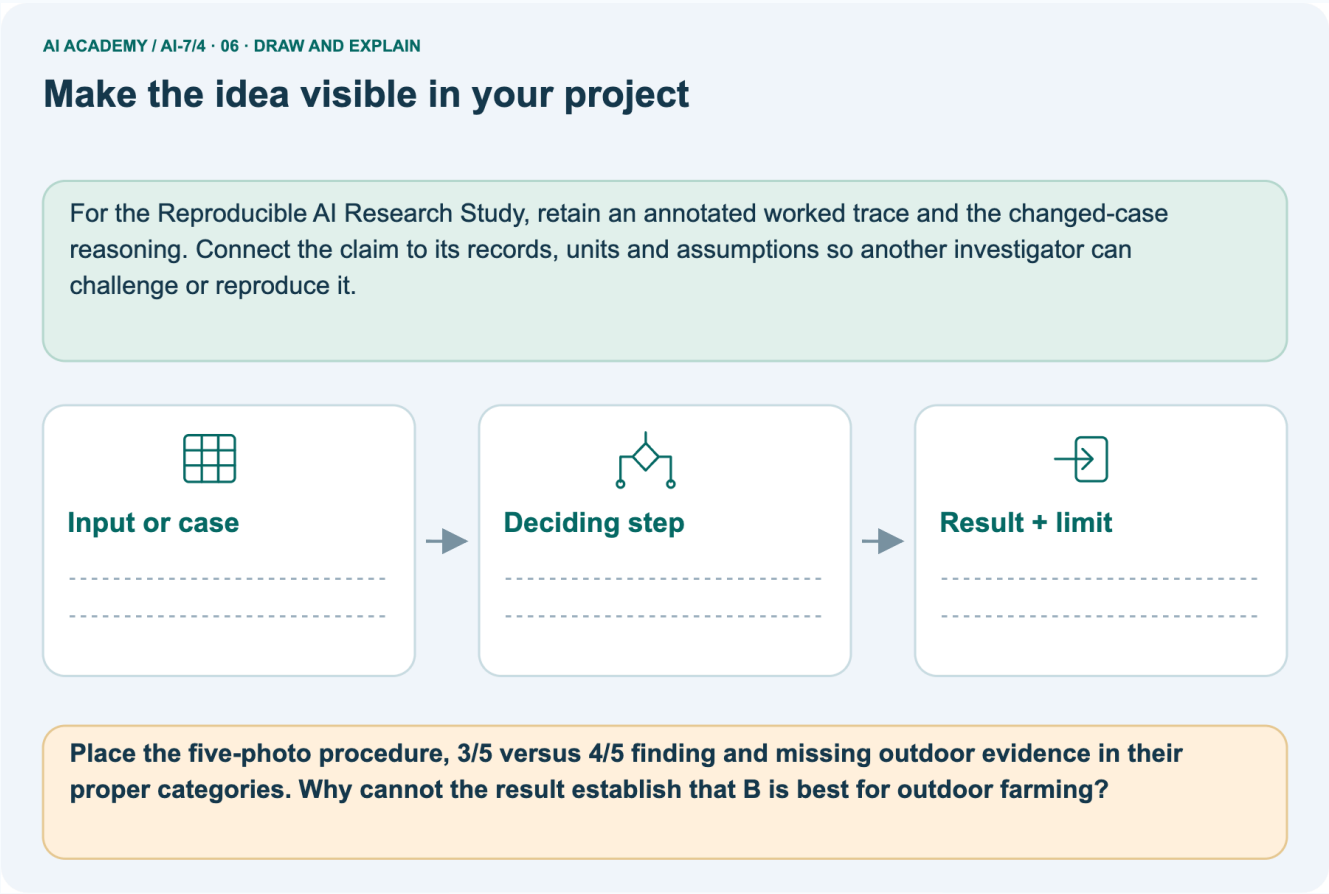
Limit: Five **observation**s cannot establish a stable ranking or causal **mechanism**; source and sampling assumptions remain important.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Five **observation**s cannot establish a stable ranking or causal **mechanism**; source and sampling assumptions remain important.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, **result** and limit. Supported question: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories.

Why cannot the **result** establish that B is best for outdoor farming?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Same five photos, labels and scoring are used for both models. A scores **60%**; B scores **80%** on this fixed set. The set supplies no outdoor-camera performance **evidence**.

Changed case: Reject that extension without outdoor **evidence**. The reported conditions do not include the proposed use population.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked **result** before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Methods results and limitations: the case

02. Follow the deciding steps

03. Two models on five shared photos

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.